STAT 654
Chapter 3: Classification

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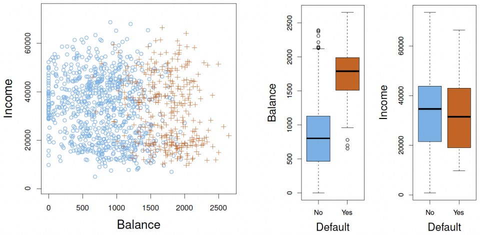
Spring 2026

An Overview of Classification

- The response variable is *qualitative*.
- For e.g., $y \in \{0, 1\}$.
- Major reasons not to perform classification using a regression method (studied earlier):
(1) A regression method cannot accommodate a qualitative response with multiple classes, and (2) A regression method will not provide meaningful estimates of $Pr(Y|X)$, even with just two classes.
- Consider $Y = 1$ if stroke; 2 if drug overdose; 3 if epileptic seizure. This coding implies an ordering on the outcomes.
- In practice there is no particular reason that this needs to be the case.
- One could choose an equally reasonable coding, $Y = 1$ if epileptic seizure; 2 if stroke; 3 if drug overdose.
- This would imply a totally different relationship among the three conditions.

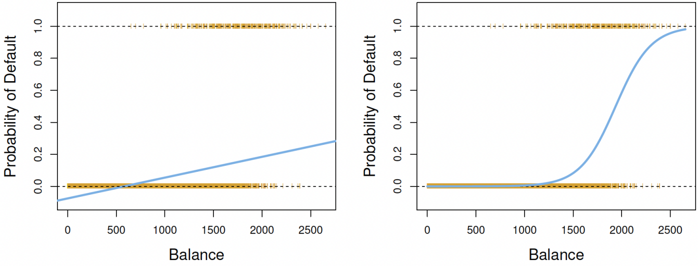
Default Data Illustration

- Data to illustrate classification: `Default` data set (ISLR2).
- Goal: Predict if an individual will default on credit card payment, on the basis of (1) annual income and (2) monthly credit card balance.



Individuals who defaulted in a given month are shown in orange, and those who did not in blue. The 'Default' variable is binary, i.e., defaulted or not.

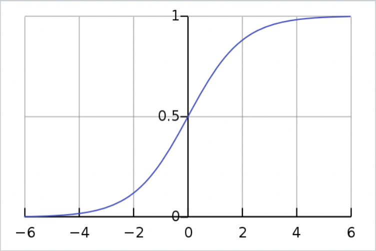
Default Data Illustration - Regression vs. Classification



Classification using the Default data. Left: Estimated probability of default using linear regression. Some estimated probabilities are negative! The orange ticks indicate the 0/1 values coded for default (No or Yes). Right: Predicted probabilities of default using logistic regression. All probabilities lie between 0 and 1.

The Logistic Model

- Model the relationship between $p(X) = Pr(Y = 1|X)$ and X.
- We model p(X) using a function that gives outputs between 0 and 1 for all values of X.
- For e.g., we use the *logistic function* (or sigmoid function) in logistic regression
  $p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$



Logistic Function

The Logistic Model

- This implies $\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$
- The quantity $\frac{p(X)}{1-p(X)}$ is called the *odds*. Can take on any value between 0 and ∞.
- This implies $log(\frac{p(X)}{1-p(X)}) = \beta_0 + \beta_1 X$. The LHS is called the *log odds* or *logit*.
- In a linear regression model, β_1 gives the average change in Y associated with a 1-unit increase in X.
- In a logistic regression model, increasing X by one unit changes the log-odds by β_1.
- Because the relationship between $p(X)$ and X is not a straight line, β_1 does not correspond to the change in p(X) associated with a one-unit increase in X.
- The amount that $p(X)$ changes due to a 1-unit change in X depends on the current value of X.

Estimating the Regression Coefficients

- The coefficients β_0 and β_1 are unknown, and must be estimated.
- In linear regression, we used the least squares approach to estimate the unknown linear regression coefficients.
- Here the method of maximum likelihood is preferred due to good statistical properties.
- Consider the *likelihood function*: $L(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{j:y_j=0}(1 - p(x_j))$
- The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to maximize this likelihood function.
- The z-statistic for $\beta_1 = \frac{\beta_1}{SE(\beta_1)}$, and so a large (absolute) value of the z-statistic indicates evidence against the null hypothesis $H_0 : \beta_1 = 0$.
- H_0 implies that $p(X) = \frac{e^{\beta_0}}{1+e^{\beta_0}}$, i.e., the probability of $\{Y = 1\}$ does not depend on X.

Making Predictions

- Think of the `Default` data.
- Once the coefficients have been estimated, we can compute the probability of default for any given credit card balance.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	-10.6513	0.3612	-29.5	<0.0001
`balance`	0.0055	0.0002	24.9	<0.0001

For the `Default` data, estimated coefficients of the logistic regression model that predicts the probability of default using balance. A 1-unit increase in balance is associated with an increase in the log odds of `default` by 0.0055 units.

Using the coefficient estimates given above, we predict that the default probability for an individual with a balance of $1000 is:
$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = 0.00576$ (plugging in values).

Qualitative Predictors

- We can use qualitative predictors with the logistic regression model using the *dummy variable* approach.
- For e.g., the `Default` dataset contains the qualitative variable `student`.
- To use student status as a predictor variable, create a dummy variable that takes value 1 for students, and 0 for non-students.

	Coefficient	Std. error	z-statistic	p-value
`Intercept`	−3.5041	0.0707	−49.55	<0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

For the `Default` data, estimated coefficients of the logistic regression model that predicts the probability of default using student status.

The coefficient associated with the dummy variable is positive, and the associated p-value is statistically significant. This indicates that students tend to have higher default probabilities than non-students.

$\hat{p}(default = 1|student = 1) = \frac{e^{-3.5+0.405X1}}{1+e^{-3.5+0.405X1}} = 0.0431.$

Multiple Logistic Regression

- Predict a binary response using multiple predictors (p predictors).
- We can generalize the logistic regression model as follows: $log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p$.
- Hence, $p(X) = \frac{e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}$
- We use the maximum likelihood method to estimate $\beta_0, \beta_1, ..., \beta_p$.

Multinomial Logistic Regression

- Classify a response variable that has more than two classes.
- For e.g., we may have three categories of medical condition in the emergency room: stroke, drug overdose, epileptic seizure.
- The logistic regression approach that we have seen only allows for K $= 2$ classes for the response variable, e.g., $Y \in \{0, 1\}$.
- It is possible to extend the two-class logistic regression approach to the setting of $K > 2$ classes $\Rightarrow$ Multinomial logistic regression.
- Select a single class to serve as the *baseline*. Without loss of generality, we select the Kth class as the baseline. Then the model is:
  $p(Y = k|X = x) = \frac{e^{\beta_{k0}+\beta_{k1}x_1+\ldots+\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\ldots+\beta_{lp}x_p}}$ for $k = 1, \ldots, K - 1$.

Multinomial Logistic Regression

- $p(Y = K|X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_{l1}x_1+\ldots+\beta_{lp}x_p}}$
- We can show that for $k = 1, ..., K-1$,
  $log\left(\frac{p(Y=k|X=x)}{p(Y=K|X=x)}\right) = \beta_{k0} + \beta_{k1}x_1 + ... + \beta_{kp}x_p$
- Once again, the log odds between any pair of classes is linear in the features.
- The decision to treat the Kth class as the baseline is unimportant.
- When classifying emergency room visits into stroke, drug overdose, and epileptic seizure, if we fit 2 multinomial logistic regression models: treating (1) stroke and (2) drug overdose, as baselines.

Multinomial Logistic Regression

- The coefficient estimates will differ between the two fitted models due to the differing choice of baseline.
- The fitted values (predictions) and the log-odds between any pair of classes will remain the same.
- Be careful with interpretation of the coefficients in a multinomial logistic regression model, since they are tied to the choice of baseline!
- E.g., setting epileptic seizure as baseline, interpret $\beta_{stroke,0}$ as the log odds of stroke versus epileptic seizure, given that $x_1 = ... = x_p = 0$.
- A 1-unit increase in X_j is associated with a $\beta_{stroke,j}$ increase in the log odds of stroke over epileptic seizure, i.e., for 1-unit increase in X_j, $\left(\frac{p(Y=stroke|X=x)}{p(Y=epilepticseizure|X=x)}\right)$ increases by $e^{\beta_{stroke,j}}$.

Probit Regression

- Binary Response $y \in \{0, 1\}$. Examples: Yes/No, Success/Failure, Disease/No Disease.
- Vector of regressors X, influencing the outcome Y. We assume that the model takes the form $p(Y = 1|X) = \Phi(X^T\beta)$.
- Here Φ is the Cumulative Distribution Function (CDF) of the standard normal distribution.
- The parameters β are usually estimated by maximum likelihood.
- We can motivate the probit model as a latent variable model.

Probit Regression

- Think of a *latent variable* Y^* as the underlying latent propensity that $Y = 1$. Note that Y^* is unobserved.
- E.g., For the binary variable, Disease/No Disease, Y^* is the propensity for Disease.
- Consider $Y^* = X^T\beta + \epsilon$, where $\epsilon \sim N(0, 1)$.
- Then Y can be viewed as an indicator for whether this latent variable is positive: Also, $Y = 1$ if $Y^* > 0$ and $Y = 0$ if $Y^* \leq 0$.
- To see that the two approaches are equivalent, note that $p(Y = 1|X) = p(Y^* > 0) = p(X^T\beta + \epsilon > 0) = p(\epsilon > -X^T\beta)$ $= p(\epsilon < X^T\beta) = \Phi(X^T\beta)$

Maximum Likelihood estimation

- Suppose data set $\{y_i, x_i\}_{i=1}^n$ contains n independent observations.
- For a single observation, we have $p(y_i = 1|x_i) = \Phi(x_i'\beta)$, and $p(y_i = 0|x_i) = 1 - \Phi(x_i'\beta)$.
- The likelihood of a single observation $\{y_i, x_i\}$ is $L(\beta; y_i, x_i) = \Phi(x_i'\beta)^{y_i}[1 - \Phi(x_i'\beta)]^{(1-y_i)})$
- The observations are independent, so the likelihood of the entire sample (joint likelihood): $L(\beta; Y, X) = \prod_{i=1}^n \Phi(x_i'\beta)^{y_i}[1 - \Phi(x_i'\beta)]^{(1-y_i)}$
- We can take the joint **log** likelihood, and maximize w.r.t. β. Thus we obtain the estimator $\hat{\beta}$, which has desirable theoretical properties.

Generalized Linear Models

- Till now, we covered linear regression (for continuous response), logistic regression and probit regression (for binary response), respectively.
- All these are part of the generalized linear models, with different *link* functions.
- Now we will consider count data.
- Consider the `Bikeshare` dataset in ISLR2.
- The response is 'bikers', the number of hourly users of a bike sharing program in Washington, DC.
- Consider predicting bikers using *mnth* (month of the year), *hr* (hour of the day, from 0 to 23), *workingday* (an indicator variable that equals 1 if it is neither a weekend nor a holiday), *temp* (the normalized temperature, in Celsius), and *weathersit* (a qualitative variable that takes on one of four possible values: clear; misty or cloudy; light rain or light snow; or heavy rain or heavy snow.)
- Treat *mnth*, *hr*, and *weathersit* as qualitative variables.

Regression with Count Data

- Consider using linear regression for data with *count* response.
- Some obvious issues appear.
- Some fitted values may be negative (predicted values may be negative).
- This calls into question our ability to perform meaningful predictions on the data.
- Raises concerns about the accuracy of the coefficient estimates, confidence intervals, and other outputs of the regression model.
- Heteroscedasticity may be observed which questions the suitability of a linear regression model.
- While the response is integer-valued, in a linear model, the response is necessarily continuous-valued. So a linear regression model is not entirely satisfactory for this dataset.

Poisson Regression

- To overcome the inadequacies of linear regression for count data, use *Poisson regression*.
- Recall the Poisson distribution: Suppose a random variable Y takes on nonnegative integer values, i.e., $Y \in \{0, 1, 2, ...\}$.
- If Y follows the Poisson distribution, then $p(Y = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, 2, ...$
- Here, $\lambda > 0$ is E(Y). Also, $\lambda = V(Y)$.
- Thus if Y follows Poisson distribution, the larger $E(Y)$, the larger is $V(Y)$.
- The Poisson distribution is typically used to model counts, since counts, like the Poisson distribution, take on nonnegative integer values.
- For regression, rather than a fixed λ, we would like to allow the mean to vary as a function of the covariates.

Poisson Regression

- Consider the model:
- $log(\lambda(X_1, ..., X_p)) = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p$
- Or equivalently, $\lambda(X_1, ..., X_p)) = e^{\beta_0 + \beta_1 X_1 + ... + \beta_p X_p}$
- To estimate the coefficients $\beta_0, \beta_1, ..., \beta_p$, we use the same maximum likelihood approach that we adopted for logistic regression.
- Specifically, given n independent observations from the Poisson regression model, the likelihood takes the form $L(\beta_0, \beta_1, ..., \beta_p) = \prod_{i=1}^{n} \frac{e^{-\lambda(x_i)} \lambda(x_i)^{y_i}}{y_i!}$, where $\lambda(x_i) = e^{\beta_0 + \beta_1 X_1 + ... + \beta_p X_p}$
- We estimate the coefficients that maximize the likelihood $L(\beta_0, \beta_1, ..., \beta_p)$, i.e., that make the observed data as likely as possible.

Poisson Regression

- Interpretation: To interpret the coefficients in the Poisson regression, note that an increase in X_j by 1 unit is associated with a change in $E(Y) = \lambda$ by a factor of e^{β_j}.
- Mean-variance relationship: Under the Poisson model, $\lambda = E(Y) = V(Y)$.
- Nonnegative fitted values: There are no negative predictions using the Poisson regression model.

Generalized Linear Models: Closing Remarks

- We have now discussed the following regression models: linear, logistic, probit and Poisson.
- All approaches use predictors $X_1, ..., X_p$ to predict a response Y. Conditional on$X_1, ..., X_p$, Y belongs to a certain family of distributions.
- Linear regression: Y $\sim$ Normal and $E(Y \mid X_1, ..., X_p) = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p$
- Logistic regression: Y $\sim$ Bernoulli and $E(Y \mid X_1, ..., X_p) = p(Y = 1 \mid X_1, ..., X_p) = \frac{e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}$

Generalized Linear Models: Closing Remarks

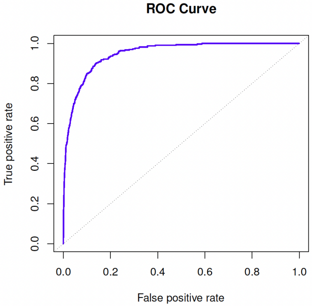
- Poisson regression: Y $\sim$ Poisson and $E(Y \mid X_1, ..., X_p) = \lambda(X_1, ..., X_p) = e^{\beta_0 + \beta_1 X_1 + ... + \beta_p X_p}$
- These equations can be expressed using a *link function* η.
- The link function applies a transformation to $E(Y \mid X_1, ..., X_p)$ so that the transformed mean is a linear function of the predictors, i.e., $\eta(E(Y \mid X_1, ..., X_p)) = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p$.
- The link functions are the following:
  linear $\eta(\mu) = \mu$,
  logistic $\eta(\mu) = log(\mu/(1-\mu))$, and
  Poisson $\eta(\mu) = log(\mu)$

Receiver Operating Characteristic (ROC) curve

- ROC curve is a graphical plot that illustrates the diagnostic ability of a binary classifier, as its discrimination threshold is varied.
- Method developed for operators of military radar receivers, hence the name.
- The ROC curve is created by plotting the *true positive rate* (TPR) against the *false positive rate* (FPR) at various thresholds.
- The true-positive rate is also known as sensitivity or recall (probability of detection).
- The false-positive rate is also known as probability of false alarm $=$ (1 - specificity).
- The performance of a classifier is given by the area under the (ROC) curve (**AUC**).

Receiver Operating Characteristic (ROC) curve

- An ideal ROC curve will be very close to the top left corner, so the larger the AUC the better the classifier.
- We expect a classifier that performs no better than chance to have an AUC of 0.5.



The ideal ROC curve is close to the top left corner, indicating a high AUC. The dotted line represents the "no information" classifier

Chapter Reference

An Introduction to Statistical Learning by G.James, D.Witten, T.Hastie, R.Tibshirani